

Six-Axis Robotic Arm – ROS 2 + Gazebo + MoveIt

This document describes how to build, launch, and verify a six-degree-of-freedom robotic arm using ROS 2 Jazzy, Gazebo simulation, `ros2_control`, and MoveIt 2.

System Overview

The robotic arm is simulated in Gazebo with full physics support. Joint control is handled by `ros2_control`, while MoveIt is used for motion planning and trajectory execution.

Requirements

• Ubuntu 22.04 • ROS 2 Jazzy • Gazebo (gz-sim) • MoveIt 2 • `ros2_control` and `ros_gz` packages

Workspace Build

```
cd ~/First_ROS2/arm_ws
colcon build
source install/setup.bash
```

Running the Simulation

Launch Gazebo and spawn the arm:
`ros2 launch arm_gazebo arm_gz.launch.py`

Verification Steps

1. Check simulation time (`/clock`) is publishing.
2. Verify controllers are active using `ros2 control list_controllers`.
3. Verify trajectory actions are available.
4. Test joint movement via `FollowJointTrajectory` action.
5. Plan and execute motions using MoveIt in RViz.

Direct Trajectory Test Example

```
ros2 action send_goal /arm_controller/follow_joint_trajectory
control_msgs/action/FollowJointTrajectory {trajectory: {joint_names: ['j1','j2','j3','j4','j5','j6'], points:
[positions: [1.2, -0.8, 1.0, -1.2, 0.7, 1.5], time_from_start: {sec: 4}}]}}
```

Important Notes

Gazebo does not publish /clock by default. The ros_gz_bridge clock bridge is required for correct execution timing. Without /clock, MoveIt execution will fail.

Conclusion

If Gazebo is running, /clock is publishing, and trajectories execute successfully, the robotic arm is fully integrated and simulation-complete.